Introduction to Limits

ANSWER KEY

Section 1: Intuitive Concept and Numerical Estimates

1 What does $\lim_{x \to c} f(x) = L$ mean intuitively?

a. **As x gets arbitrarily close to c (from either side, $x \neq c$), $f(x)$ gets arbitrarily close to L** ✓

b. $f(c) = L$ always

c. The slope of f at c is L

d. $f(x)$ is continuous at c

A limit describes the value that $f(x)$ approaches as x nears c, completely independent of what happens at $x = c$ itself.

2 Use numerical estimation with $x = 1.9, 1.99, 2.01, 2.1$ to estimate $\lim_{x \to 2} (3x - 1)$.

a. 5 ✓

b. 6

c. 4

d. 3

$3(1.99) - 1 = 4.97$; $3(2.01) - 1 = 5.03$. Values converge toward 5 as $x \to 2$.

3 If $f(3)$ is undefined, can $\lim_{x \to 3} f(x)$ still exist?

a. **Yes, because limits describe behavior near $x = 3$, not at $x = 3$** ✓

b. No, limits require $f(c)$ to be defined

c. Only if $f(x) = 0$

d. Only for polynomials

The existence of a limit at c depends entirely on values in a punctured neighborhood around c, regardless of whether $f(c)$ is defined.

Section 2: Graphical Limits and One-Sided Limits

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4 When does the two-sided limit $\lim_{x \to c} f(x)$ exist?

a. **If and only if both one-sided limits exist and are equal ($\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$)** ✓

b. Whenever $f(c)$ exists

c. Whenever $\lim_{x \to c^+} f(x)$ exists

d. If f is defined on $[c, \infty)$

The two-sided limit exists if and only if both the left-hand limit and the right-hand limit exist and agree on a common value L.

5 For the step function $f(x)$ defined as 2 for $x < 1$ and 5 for $x \geq 1$, find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$.

a. **Left limit $= 2$, Right limit $= 5$** ✓

b. Left limit $= 5$, Right limit $= 2$

c. Both equal 2

d. Both equal 5

As $x \to 1^-$ (from the left), $f(x) = 2$. As $x \to 1^+$ (from the right), $f(x) = 5$.

6 Does the two-sided limit $\lim_{x \to 1} f(x)$ exist for the step function in the previous question?

a. **No, because the left limit (2) does not equal the right limit (5)** ✓

b. Yes, it equals 5

c. Yes, it equals 3.5

d. Yes, it equals 2

Because $\lim_{x \to 1^-} f(x) = 2 \neq 5 = \lim_{x \to 1^+} f(x)$, the two-sided limit does not exist (DNE).

Section 3: Algebraic Evaluation of Limits

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7 Evaluate $\lim_{x\to 4} \frac{x^2-16}{x-4}$.

a. 8 ✓

b. 0

c. 4

d. Does not exist

Indeterminate form $0/0$. Factor: $\frac{(x-4)(x+4)}{x-4} = x+4$. As $x \to 4$, $4+4=8$.

8 Evaluate $\lim_{x\to 0} \frac{\sqrt{x+9}-3}{x}$.

a. 1/6 ✓

b. 1/3

c. 0

d. 1/9

Multiply numerator and denominator by conjugate $(\sqrt{x+9}+3)$: $\frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \frac{1}{\sqrt{x+9}+3} \to \frac{1}{3+3} = \frac{1}{6}$.

9 Evaluate $\lim_{x\to -2} \frac{x^3+8}{x+2}$.

a. 12 ✓

b. 0

c. 4

d. 8

Sum of cubes: $\frac{(x+2)(x^2-2x+4)}{x+2} = x^2-2x+4$. At $x=-2$: $(-2)^2-2(-2)+4 = 4+4+4 = 12$.

Section 4: Limits Involving Infinity

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10 Evaluate $\lim_{x\to\infty} \frac{5x^2 - 2x}{2x^2 + 7x - 1}$.

a. $5/2$ ✓

b. ∞

c. 0

d. -2

Divide by x^2: $\lim_{x\to\infty} \frac{5 - 2/x}{2 + 7/x - 1/x^2} = \frac{5}{2}$.

11 Evaluate $\lim_{x\to\infty} \frac{3x + 1}{x^2 + 5}$.

a. 0 ✓

b. 3

c. ∞

d. $1/5$

The degree of the denominator is higher than that of the numerator, so the limit as $x \to \infty$ is 0.

12 Evaluate $\lim_{x\to 3^+} \frac{1}{x-3}$ and $\lim_{x\to 3^-} \frac{1}{x-3}$.

a. $+\infty$ **and** $-\infty$ ✓

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 0 and 0

As $x \to 3^+$, $x - 3 > 0$, so $1/(x-3) \to +\infty$. As $x \to 3^-$, $x - 3 < 0$, so $1/(x-3) \to -\infty$.

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